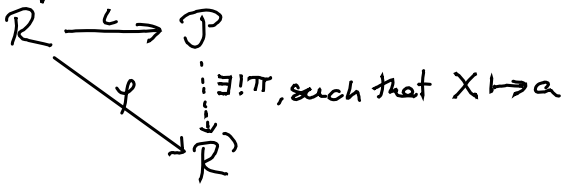
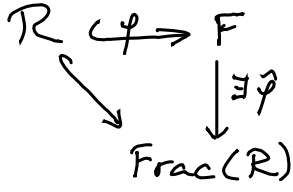
- Ring R : $\cdot (R,+)$ abelian
  $\cdot (R,\cdot)$ monoid
  $\cdot$ distributive law
- $\frac{R/I}{J/I} \cong \frac{R}{J}$
- $\ker(eval_a) = \langle X-a \rangle$, $\frac{P}{\ker(eval_a)} \cong R$
- e,e' complementary idempotents: $R \cong Re \times Re'$, $Re \cong \frac{R}{Re'}$
- multiplicative set S: $1 \in S \wedge S \cdot S \subseteq S$
- $P \lhd R$ is prime if $R-P$ is multiplicative
  $(\Leftrightarrow (ab \in P \Rightarrow a \in P \vee b \in P))$
- integral domain is a Ring with no zero devisors
  ($\langle 0 \rangle$ is prime)
- UL for polinoms



- R is integral domain $\Leftrightarrow$ R is subring of a field
- R is integral domain $\Rightarrow$ we can construct $frac(R)$
- UL for $frac(R)$



- J prime $\Rightarrow$ $\varphi^*(J)$ prime
- I prime, φ surj $\Rightarrow$ $\varphi_*(I)$ prime
- $P \trianglelefteq R$ is prime $\Leftrightarrow$ R/P is integral domain
- R is a field $\Leftrightarrow$ $\langle 0 \rangle$ is max ideal
- $M \trianglelefteq R$ max ideal $\Leftrightarrow$ R/M is a field
- M max ideal $\Rightarrow$ M prime ideal
- every proper ideal is contained in max ideal
- $x \in R^\times \Leftrightarrow x \notin$ max ideal
- a is irreducable $\Leftrightarrow$ a has no proper devisors $\Leftrightarrow$
  $\Leftrightarrow \langle a \rangle$ is max among principal ideals
- p is prime element $\Leftrightarrow p \notin R^\times \wedge (p \mid ab \Rightarrow p \mid a \vee p \mid b)$
  $\Leftrightarrow \langle p \rangle$ is prime ideal
- unique factorisation domain (UFD)
  1) $\forall a \in R - \{0\}$ is a product of irreducable elem. or units
  2) factorisation is unique up to order and units
- R is int domain $\Rightarrow$ (R is UFD $\Leftrightarrow$
  1) every chain of principle ideals $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \ldots$ stabilizes
  2) pirredecable $\Rightarrow$ p prime)
- principal ideal domain (PID) if every ideal is principal

- R PID $\Rightarrow$
  - p irreducable $\Rightarrow$ p prime
  - $I \neq \langle 0 \rangle \trianglelefteq R \Rightarrow I$ is max
  - R is UFD
- Jacobson radical $\operatorname{rad}(R) = \bigcap_{M \trianglelefteq_{max} R} M$
- R UFD $\Rightarrow R[x]$ UFD
- 3. izrek o izomorfizmu $\frac{R/(a_1 \ldots a_n)}{(b_1 \ldots b_m)} = \frac{R}{a_1 \ldots a_n b_1 \ldots b_m}$

  Izreki o izomorfizmih:
  1) $R/\ker\varphi \cong \operatorname{Im}\varphi$
  2) $\frac{K+I}{I} \cong K/K \cap I$
  3) $\frac{R/I}{J/I} \cong R/J$